ADVANCED DATA ANALYSIS AND MACHINE LEARNING

NASA TURBOFAN: REMAINING USEFUL LIFE ESTIMATION OPTIMIZATION

LUT University

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https://github.com/aminhm/NASA-Turbofan-A2.git

Content

1 Communication channel and code sharing strategy

The group will use Teams as a primary communication channel and GitHub will be used as a coding collaboration platform. Python was decided to be used as a tool to research the dataset.

2 Dataset description and identification of challenges of the data

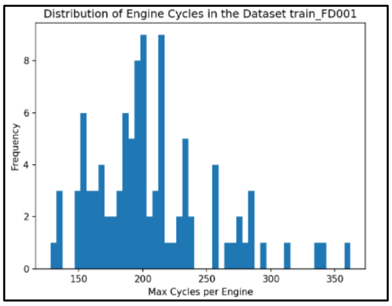

Figure 1. Distribution of engine cycles in the dataset train_FD001.

The dataset used in this study is the NASA Turbofan Jet Engine Degradation Dataset. The dataset consists of four different datasets: FD001, FD002, FD003 and FD004. Datasets contain multiple multivariate time series, each representing the operational history of a different engine.

As expected, each engine has unequal amount of data. For example, distribution of maximum engine cycles in the dataset train_FD001 is shown in figure 1.

There seems to be no missing or extra values in the data, however datasets FD001 and FD003 have some columns with constant values. Physical meaning for the operational settings and sensor measurements is not specified.

3 Visualization and comment on the dataset

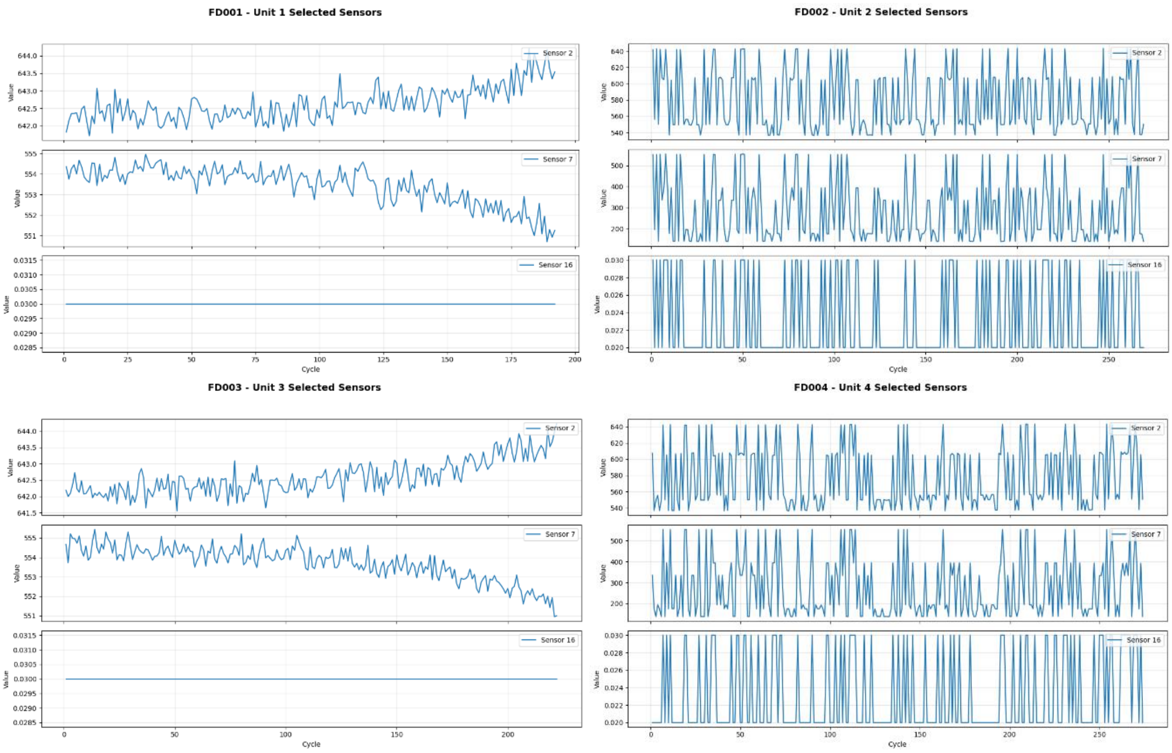

Figure 2. Distribution of variables in all datasets

The training datasets consist of 26 variables including unit number, time in cycles, three operational settings, and 21 sensor measurements. The number of observations is 20,631 for the FD001, 53759 for the FD002, 24720 for the FD003, and 61249 for the FD004. According to data, operational settings show distinct patterns. Setting 3 is nearly constant, setting 1 is normal in FD001 and FD003 but multimodal in FD002 and FD004, and setting 2 is multimodal depending on the system. Sensor measurements range from constant to normal and multimodal. They also reflect diverse engine states.

As multivariate time series, the data captures each engine's evolution over time, and the 'time, in cycles' variable increases monotonic over the data, so it is a time-series problem. Fig 2 shows the constant and different patterns of sensor 2,7, and 16 data across 'time, in cycles'.

4 Exploratory data analysis with PCA

Principal Component Analysis (PCA) was applied to the standardized matrixes of the datasets to analyse the correlations between sensors' data and retrieve the principal components that capture most of the variance in the dataset. In the scree plots in Figure 3, the first two components are the most impactful on the variance of the dataset, contributing to ~ 65% in FD001, ~ 67% in FD003 and ~ 97% in both FD002 and FD004.

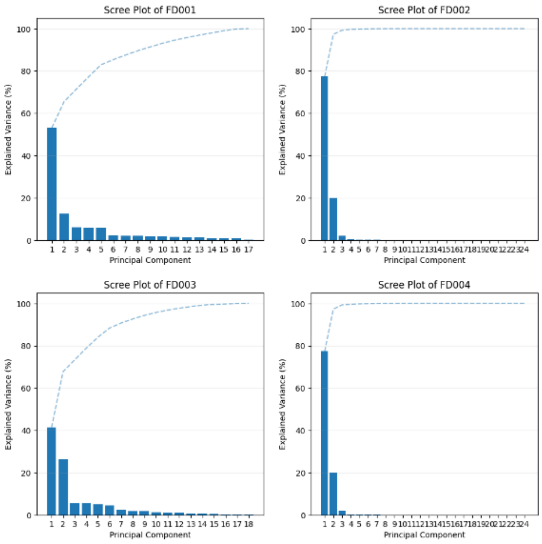

Figure 3. Scree plot of datasets with explained and cumulative variance per principal component.

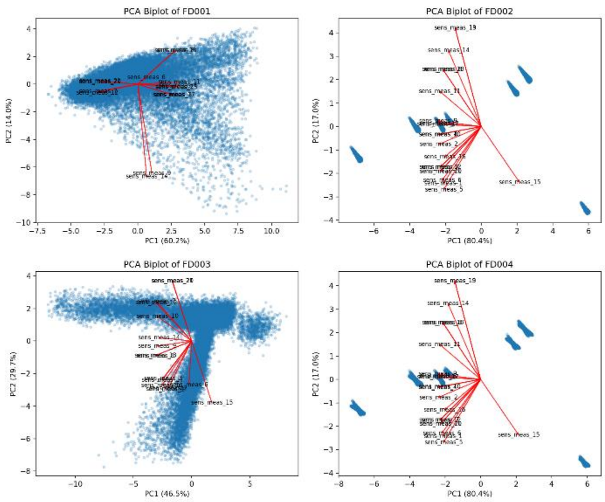

Figure 4. Biplots of data projected onto PC1 and PC2 and loadings for each sensor.

The loadings bars in Figure 5 show how each sensor relates to a principal component. Large positive or negative bars mean a strong link to that component. Bars close to zero indicate the sensor contributes little new information.

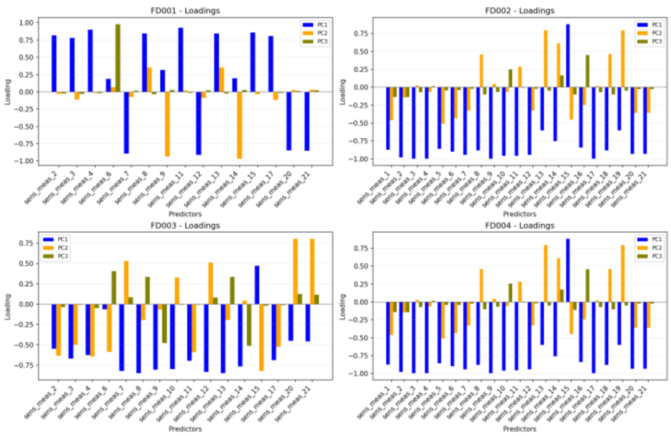

Figure 5. Loadings bar plots to analyse correlations between principal components and sensors

By looking at the plot, it is possible to determine that PC1 has a positive correlation with most sensors, while PC2 and PC3 focus attention on the differences between sensor groups. This shows that using the first three components is sufficient for further analysis.

5 Data pretreatment

In this stage, non-informative variables were removed from each dataset. The three operational settings were excluded in all cases, as well as sensors that remain constant or near constant. Specifically, FD001 and FD003 had sensors 1, 5, 6, 10, 16, 18, and 19 removed, while FD002 and FD004 had sensors 13, 16, and 19 removed.

The data is centered and scaled using Standard Scaler. It subtracts the mean from each sensor data, and the mean becomes zero. Then the data is divided by the standard deviation and standard deviation becomes zero. The formula of Standard Scaler is shown in formula 1.

$$x_{standardized} = \frac{x \ -\mu}{\sigma} \tag{1}$$

Where $x_{standardized}$ is standardized data, x is original data, μ is mean of the data and σ is standard deviation of the data.

There are no missing values in the datasets, however, all four datasets (FD001–FD004) contain no missing values. Extreme values were found in 14 sensors for FD001–FD003 and 15 sensors for FD004, which is expected in run-to-failure engine data.

In addition, there is no misalignment in datasets which indicates that there are no missing cycles for each unit, and the original data is the same as the synchronized data.

6 Visualization of pretreated data

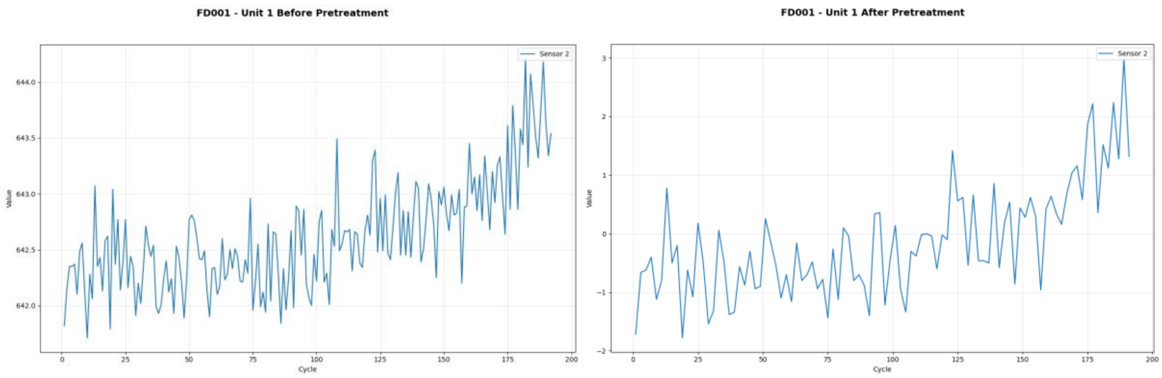

Figure 6. The data of the second sensor of FD001 dataset before and after pretreatment.

Figure 6 shows the data of the second sensor of FD001 dataset before and after pretreatment as an example.

7 Modelling goal

The modelling goal is to recreate a regression model, removing redundant sensors from the base regression model and keeping just the most important ones, capable of calculating an estimate of the Remaining Useful Life (RUL) of the engine calculated with all sensors.

For this purpose, four different training data were given for the analysis and recreation of the model. The best division strategy found is to split the units into three different sections:

- one for training / calibration (60% of the units),
- one for validation (20% of the units),
- one for testing (20% of the units).

8 Model calibration strategy

For training / calibration purposes, 60% of the total units are analyzed. Before the calibration step, data pretreatment was implemented to normalize data (z-score approach), remove sensors that measure constant data (not useful for the main goal) and handle missing values (not present after removing constant sensors). The main purpose of calibration is to estimate the model parameters and help identifying which sensors are more important for predicting the RUL.

Since, as seen in the scree plot in Figure 3, just two principal components are sufficient for describing most of the variance, it indicates that many sensors are highly correlated, and so the main strategy for this step is to use Partial Least Squares (PLS), as it efficiently reduces dimensionality and handles multicollinearity among the sensor readings.

9 Model validation strategy

To evaluate the correctness of the parameters' estimation calculated in the training part, a validation analysis is performed using a separate set of units (20% of the total units) as a validation set.

For the evaluation, the main strategy is to apply a k-fold cross-validation procedure to the calibration data, to minimize the Root Mean Squared Error (RMSE) or the Predicted Residual Sum of Squares (PRESS), and to maximize the cross-validated coefficient of determination Q^2_{CV}. The validation output is then used as feedback to help adjust model parameters during training or sensor selection.

10 Model testing strategy

After a suitable subset of sensors has been found, the testing part begins. The remaining 20% of the total dataset is used to provide an unbiased estimate of predictive performance.

To assess the accuracy of the computation, metrics such as RMSE and R^2 can be computed to ensure that the sensors subset is sufficient for estimating the exact RUL calculated using all sensors, to address any trade-off between sensor reduction and prediction quality. Also, Q^2_{test} is useful with R^2 of the calibration to identify if the model is overfitted (low Q^2_{test} with high R^2).

11 Mathematical methods used in modelling

The group will be using partial least squares (PLS) regression as a model to solve the task.

According to the Handbook of Partial Least Squares, Partial Least Squares (PLS) is a statistical method used to model complex relationships between multiple observed and latent variables. It is a component-based approach, where causality is expressed through linear conditional expectations (Esposito Vinzi et al., 2022; Wold, 1975).

The development of a PLS model begins with a training dataset consisting of N observations (such as cases, objects, compounds, or time points). Each observation includes K independent variables, denoted as x_k (where k = 1, ..., K), and M dependent variables, denoted as y_m (where m = 1, ..., M). The data is structured into two matrices: X with dimensions (N × K) and Y with dimensions (N × M).

After the model is constructed, it can be used to make predictions for new observations based on their X-values. The output includes predicted X-scores, residuals for the X variables, the standard deviations of these residuals, and predicted Y-values with associated confidence intervals (Wold et al., 2001).

According to lecture materials, the model predictions $\hat{Y}$ can be expressed in a following way shown in formula 2. (Reinikainen, 2025)

$$\hat{Y} = X(W(P^T W)^{-1} Q^T) \tag{2}$$

where X is X matrix (input data), W is Weight matrix, P is Orthogonal loading matrix and Q is Orthogonal loading matrix .

12 Modeling flow chart

Initial proposition for the modeling flow chart is shown in Figure 7. In this case the starting point of the modeling process was when we acquired the cleaned data. In total, flow chart has 8 steps, and the last step is reporting the results.

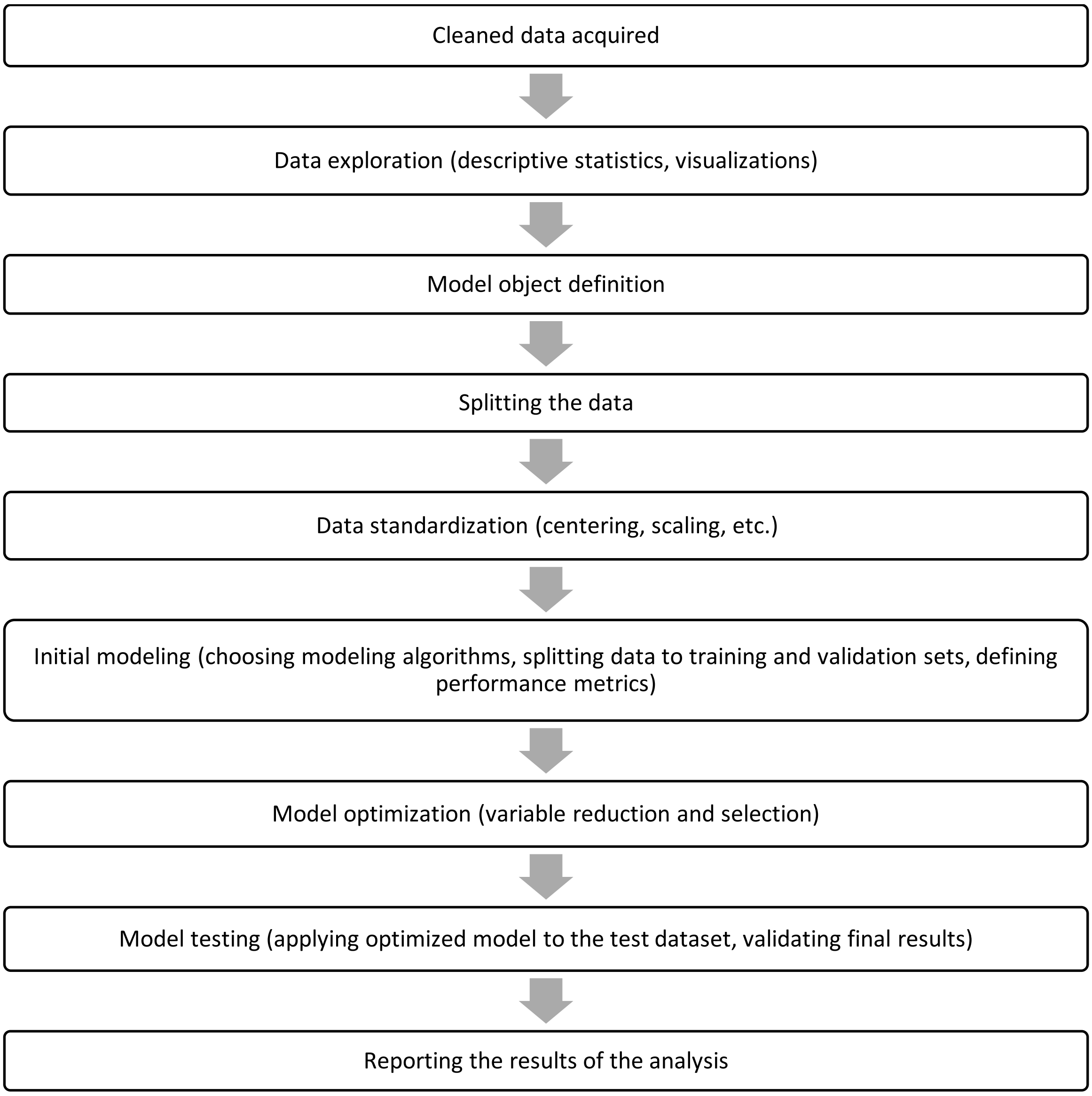

Figure 7. Proposed modelling flow chart.

13 Calibrating the model and achieving the modeling goal

The modeling goal was to reduce the model using only sensors that contribute most in terms of predicting remaining useful life of the engine. First, an analysis using PRESS and Q^2 scores was conducted, and it was found that the optimal number of components for prediction is 2. The PRESS and Q^2 scores versus number of PLS components are shown in Figure 8.

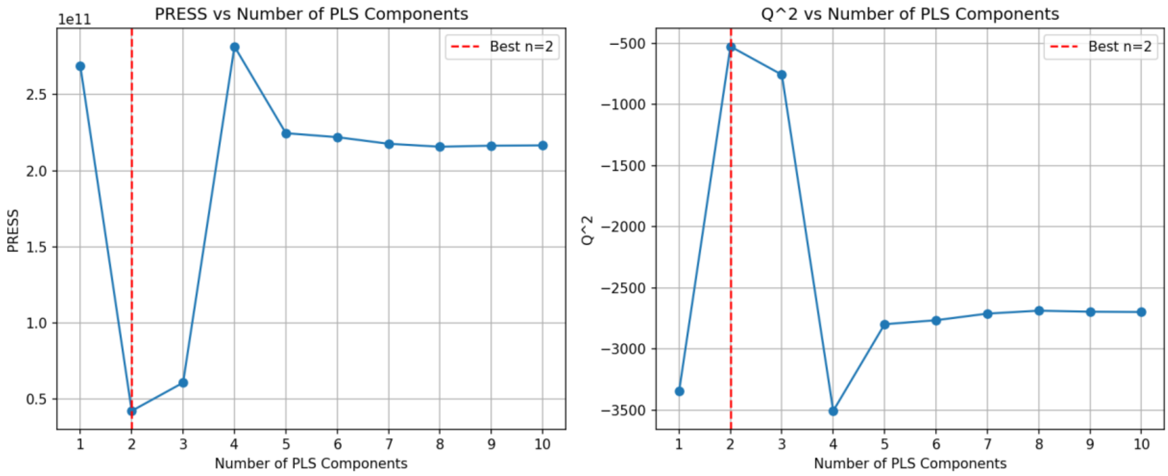

Figure 8. PRESS and Q^2 values versus number of PLS components in the model.

Variable importance in projection is shown in Figure 9. According to the results, the most useful sensors that have a VIP score greater than 1 (used as threshold) are only 10, with sensor measurements 20 and 21 having the highest VIP.

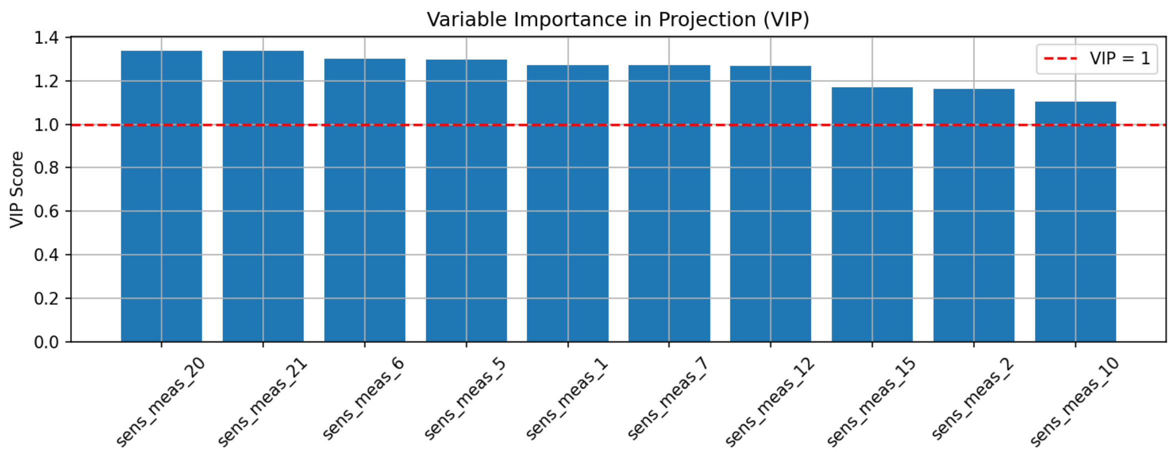

Figure 9. Model variable importance in projection.

The model validation results are shown in Figure 10 and model testing results are shown in Figure 11. The calculated RMSE values for validation data were 74.5 and for test data 75.7.

Q2 score in validation phase is 0.298, while in testing phase is 0.292. This indicates that the model explains about 30% of the variance in the true RUL values on unseen data. So, the selected sensors capture relevant information, but improvements can be made.

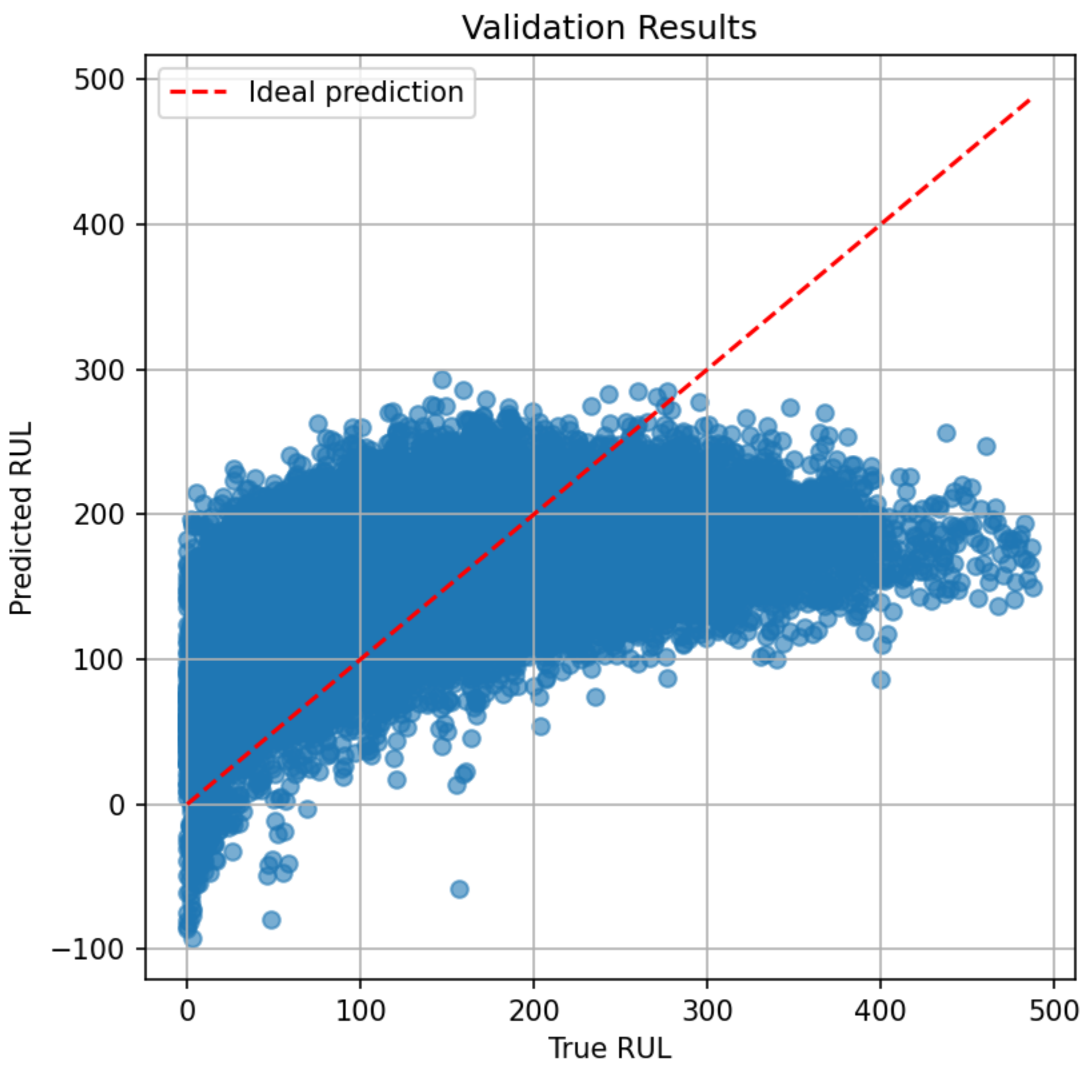

Figure 10. Model validation results. Predicted RUL is shown in the y-axis and measured RUL is shown in the x-axis.

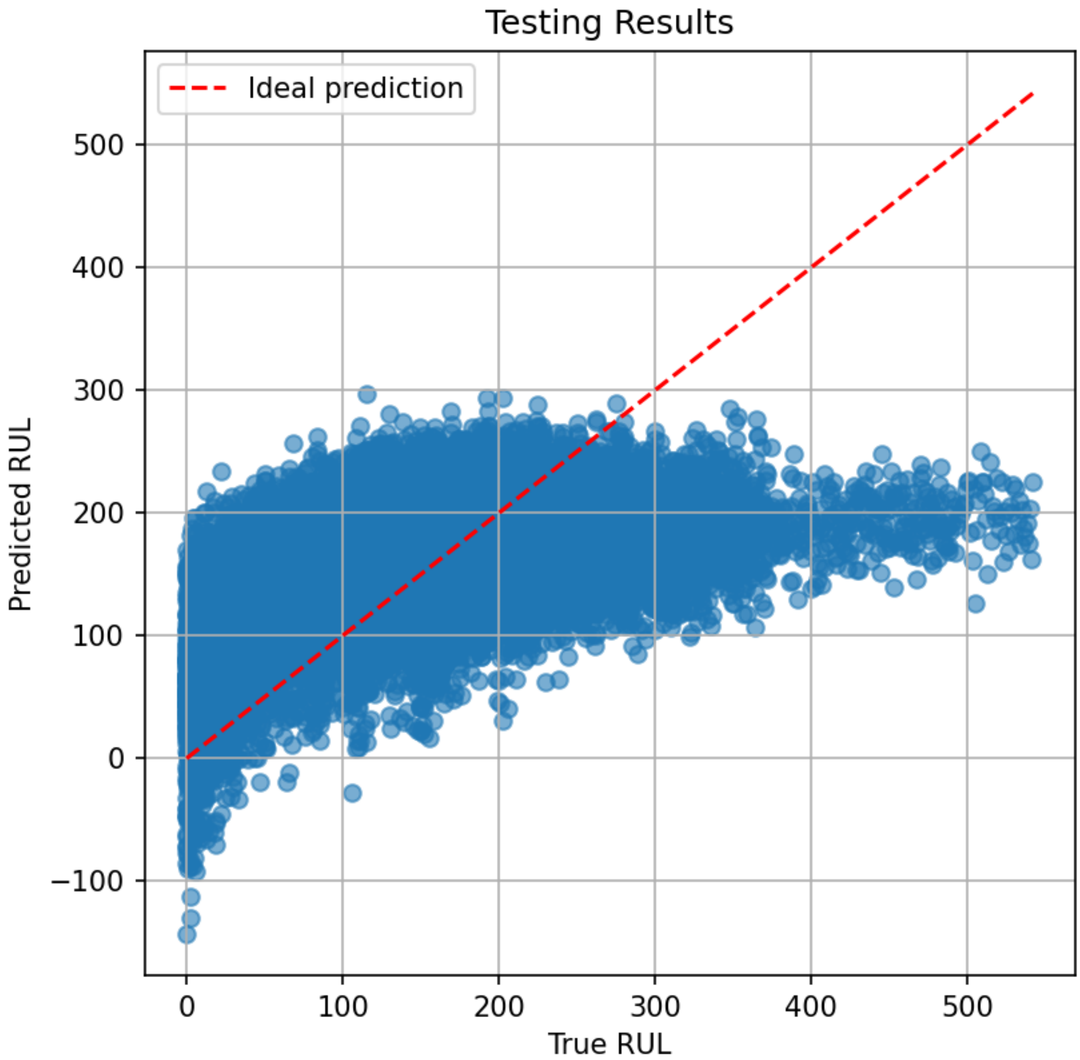

Figure 11. Model testing results. Predicted RUL is shown in the y-axis and measured RUL is shown in the x-axis.

14 Conclusion

After initial review of the results, it can be seen that there is still some room for improvement, which will be done in the upcoming weeks before the final submission of this report.

15 References

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